

Urinalysis & Urine Culture - Summary Table

PAS 673: Introduction to Clinical Year II - Winter 2026

Indications for Urinalysis

Indicated	Not Indicated
Signs/symptoms of acute infection Reduction of GFR (acute or chronic) / AKI Check for albuminuria Suspected kidney disease (PE findings, comorbidities) Monitoring / progression / staging of CKD	Asymptomatic screening before urologic procedure

Gross Assessment

Finding	Normal	Causes / Differential
Color	Clear, light yellow	Red to brown: hemoglobin, myoglobin, RBCs Medications: rifampin, phenytoin Foods: beets, rhubarb, food dyes
Turbidity	Clear	Infection, genital secretions, crystals, cells
Odor	Mild / aromatic	Pungent / ammonia: infection, retention Fruity: ketones (DKA, starvation)

Urine Dipstick Parameters

Parameter	Normal	Increased / Positive	Decreased / Negative	False Pos / False Neg
Specific Gravity (SG)	1.005 - 1.030	Dehydration; glucose in urine; heart failure; hyponatremia; renal artery stenosis; SIADH	Excessive water intake / psychogenic polydipsia; renal tubular necrosis; diabetes insipidus; kidney failure; hyponatremia; pyelonephritis	-
pH	4.6 - 8.0	Increased (basic): UTI; vomiting / gastric suctioning; renal tubular acidosis (some types)	Decreased (acidic): metabolic acidosis; DKA; starvation; diarrhea; kidney failure	-
Heme	Negative (trace / small / mod / large)	Bladder cancer; kidney stone; UTI; glomerulonephritis; trauma; rhabdomyolysis; exercise	-	FP: menses, dehydration, semen Confirm hematuria with microscopy
Leukocyte Esterase	Negative (1+, 2+, 3+)	UTI (marker of WBCs from lysed neutrophils/macrophages); microscopy recommended	-	FP: contamination, STI FN: very concentrated urine, proteinuria, glucosuria, ketonuria
Nitrite	Negative	UTI (Enterobacteria convert nitrate → nitrite via nitrate reductase)	-	FN: Enterococcal species (do not reduce nitrate); dilute urine; insufficient bladder dwell time
Protein (albumin)	Negative Mod albuminuria 30-300 mg Severe >300 mg	Persistent: primary glomerular (IgA/IgM, MCD, FSGS, MPGN); secondary (SLE, DM, NSAIDs/ACEi, HIV/syphilis/post-strep, malignancy, sarcoid, sickle cell); tubular (heavy metals, interstitial nephritis); overflow (myeloma, myoglobinuria) Transient: CHF, dehydration, exercise, fever, seizure	-	Dilute urine underestimates; concentrated overestimates. Confirm with ACR, 24-hr urine, sediment, UPEP, renal function
Glucose	Negative	Elevated blood glucose: Diabetes mellitus (glycosuria typically >180 mg/dL plasma) Normal blood glucose: Fanconi syndrome; renal glycosuria (proximal tubule defect)	-	-
Ketones	Negative	DKA; hyperemesis; starvation; dehydration; low-carb diets	-	Products of fat metabolism, not normally in urine
Bilirubin	Negative	Conjugated hyperbilirubinemia: liver dysfunction, biliary obstruction	-	-
Urobilinogen	0.1 - 1.8 mg/dL (>2.0 elevated)	Hemolysis / hemolytic anemia; hepatocellular disease; biliary tract issues	Antibiotic suppression of gut bacteria; bile duct obstruction	End product of conjugated bilirubin metabolized by gut bacteria

Microscopic Examination

Centrifuge 10-15 mL fresh urine; transfer one drop to slide with cover slip. Identify casts, cells, crystals, and bacteria.

Element	Normal	Significance / Causes
Erythrocytes (RBCs)	0-3 RBCs / HPF	Confirm with dipstick heme. Transient vs persistent. Glomerular vs nonglomerular causes - assess morphology (dysmorphic = glomerular)
Leukocytes (WBCs)	Men: <2 WBC/HPF Women: <5 WBC/HPF	Pyuria - infection, inflammation; correlate with LE and nitrite
Epithelial cells	Few	Squamous: contamination Renal tubular: tubular damage
Casts	Hyaline casts may be normal	Cylindrical shape reflects tubule of formation; localizes disease to a specific GU site (e.g., WBC casts → upper tract / pyelonephritis)
Crystals	Variable	Type depends on urine pH and underlying disease (uric acid, calcium oxalate, struvite, cystine)
Bacteria	None	Gram-negative rods, streptococci, staphylococci - correlate with culture

When to Examine the Urine Sediment

Should examine	Could examine	Unnecessary
Clarify findings on dipstick AKI Initial work-up of undifferentiated CKD	Follow-up of acute and chronic kidney disease Initial evaluation of kidney stones	Negative dipstick + mildly impaired renal function DM with mildly impaired kidney function and albuminuria

Urine Culture & Sensitivity

Topic	Detail
Positive culture	>100,000 CFU/mL = positive. 1,000-100,000 CFU/mL may indicate UTI in symptomatic patients.
MIC	Minimum Inhibitory Concentration - lowest antibiotic concentration that prevents visible bacterial growth
Appropriate uses	Symptoms suggestive of UTI (<i>except healthy nonpregnant young females with typical acute simple cystitis</i>) New onset / worsening sepsis without another source Fever or altered mental status without another source
Inappropriate uses	Odorous, cloudy, or discolored urine without other localizing signs/symptoms Reflex cultures based on UA findings (e.g., pyuria) without other indications Documenting response to therapy (unless symptoms fail to resolve)

Case Summary

Case	Patient	Key Findings	Diagnosis / Management
#1	22F, dysuria, urgency, frequency, suprapubic pressure; no fever / CVA tenderness	Cloudy urine; LE ++; nitrite +; 15-20 WBCs/hpf; moderate bacteria	Acute uncomplicated cystitis Empiric outpatient therapy; culture often not required in healthy nonpregnant young female with typical symptoms
#2	75M with BPH (tamsulosin), worsening hesitancy, dribbling, fever 100.8°F, dark/foul urine	Turbid; LE +++; nitrite +; >50 WBCs/hpf; many bacteria; PVR 250 mL; WBC 12,500	Complicated UTI (male + obstruction/retention) Urine culture; broad-spectrum antibiotics; address retention
#3	26F, 22w gestation; chills, N/V, R flank pain; T 102.4°F; tachycardic; +CVA tenderness	Turbid 'meat-wash'; LE +++; nitrite +; packed WBCs; WBC casts ; gram-neg rods; WBC 16k w/ left shift	Acute pyelonephritis in pregnancy Admit; IV antibiotics; fetal monitoring; culture-guided therapy
#4	24F, 12w gestation, asymptomatic; routine prenatal screening	Culture: 100,000 CFU/mL E. coli; LE trace; nitrite +; 5-10 WBCs/hpf	Asymptomatic bacteriuria in pregnancy <i>Treat</i> (unlike non-pregnant adults) to reduce risk of pyelonephritis and preterm labor